

TEJAS PANDYA

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EDUCATION

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	Expected 05/27
Master of Science in Financial Engineering GPA: 3.8/4.0	
Graduate Teaching Assistant: Financial Economics; C++ Financial Software Laboratory; Value Investing	
SVKM'S NMIMS, SCHOOL OF BUSINESS MANAGEMENT , Mumbai, India	04/22
Master of Business Administration, Concentration in Finance GPA: 3.3/4.0	
VELLORE INSTITUTE OF TECHNOLOGY , Vellore, India	06/20
Bachelor of Technology in Mechanical Engineering GPA: 8.9/10.0	

PROGRAMMING / TECHNICAL SKILLS / CERTIFICATIONS

Programming: C/C++ (C++20, STL, Boost), Python, SQL, Bash, LaTeX/Typst
Libraries & Frameworks: NumPy, Pandas, Polars, SciPy, scikit-learn, statsmodels, LightGBM, XGBoost, PyTorch, pytest
Data & Tools: Linux/Unix, Git, CMake, Streamlit, Shiny, Excel, Bloomberg Terminal, Tableau, Power BI
Certifications: CFA Level I; C++ Programming for Financial Engineering (Baruch College); Bloomberg Market Concepts; CFI FMVA

COURSEWORK HIGHLIGHTS

Quantitative Methods & Computing: Financial Econometrics, Machine Learning in Financial Engineering, Quantitative Methods in Finance, Financial Computing, Applied Numerical Methods, Data Structures and Algorithms
Risk, Valuation & Derivatives: Financial Risk Management, Valuation for Financial Engineering, Quantitative Portfolio Management, Options Pricing & Stochastic Calculus, Fixed Income Securities and Interest Rate Derivatives

EXPERIENCE

NEW YORK UNIVERSITY, TANDON SCHOOL OF ENGINEERING , Brooklyn, NY	05/26 - 08/26
<i>Quantitative Researcher, QWIM Industry Capstone</i>	
- Implemented a log-periodic power-law singularity bubble-confidence tilt as the fourth ablation level above regime-conditioned covariance blending and quantile-regression tail estimation, the strongest of eight variants, holding Sharpe 0.56 against 0.50 on the 2008-2025 window - Diagnosed thirteen environment-configuration defects behind a 278-file team pytest suite that reported success while executing zero tests on macOS, restoring 10,423 tests to green and documenting the 22 pre-existing failures rather than leaving them quietly reported as green - Built the Regime Risk Budgeting dashboard tab and its Typst client-report exporter, 1,058 lines of Python across ten modules with twelve unit tests, written fail-safe so that any missing artifact degrades only that one report section rather than breaking the whole client report	
NOMURA SERVICES INDIA PRIVATE LIMITED , Mumbai, India	08/22 - 07/25
<i>Senior Risk Analyst, Market Risk Infrastructure</i>	
- Owned the lifecycle of VaR and sensitivity models across equities, rates, FX and commodities, from daily production runs through backtesting; Python anomaly detection triaged exceptions to stale marks, erroneous data or booking errors, cutting investigation turnaround 40% and restoring SLAs - Delivered quarterly Basel 2.5 and FRTB stress-testing and capital submissions alongside front-office and regulatory teams, engineering rules-based validation over delta, vega and curvature sensitivities that reconciled fully against front-office pricing and caught anomalies pre-submission - Engineered Python and SQL pipelines across equities, rates, FX and commodities that aggregate, validate and reconcile daily P&L, positions and Level-1/Level-2 market data into the VaR and sensitivity dashboards, eliminating more than 50 hours of manual processing every month - Led a three-person workstream on US-region end-of-day risk reporting, cutting overnight batch runtime 35% through SQL query optimisation and job scheduling, and built the VaR, P&L and exception dashboards that senior stakeholders read each morning before the US open	

ACADEMIC PROJECTS

TREASURY-CURVE SWAP VALUATION & HISTORICAL COUNTERPARTY RISK	05/26
<i>Python, NumPy, SciPy</i> github.com/tejaspandya9598/sofr-swap-pnl-attribution	
- Priced historical counterparty CVA from 20,000 paired exposure paths over 8,478 days of joint rate and credit history, measuring wrong-way risk at \$89,553, 16.9% above a decoupled baseline sharing its curve paths bit-for-bit, at a \$2.35M peak expected exposure, with no collateral agreement - Simulated curves inside PCA factor space, three components covering 98% of curve variance, after unconstrained pillar walks reached 740bp of 2s30s against the 401bp maximum the market has printed and produced curves carrying no positive discount factor at the long end	
HFT LIMIT ORDER BOOK & MATCHING ENGINE	02/26
<i>C++20, CMake, CTest</i> github.com/tejaspandya9598/hft-matching-engine	
- Built a deterministic price-time-priority matching engine timing roughly 21M messages per second on an Apple M2 over a pre-generated synthetic message mix, and 10.7M through the lock-free end-to-end pipeline, on a zero-allocation hot path with 64-byte cache-aligned structures - Reached O(1) average add, cancel and execute through intrusive per-level lists with hash lookup, a new price level costing O(log L) for best-bid tracking; 20 CTest assertions plus five benchmark checks pin price-time priority, partial fills and pool reuse under 10k-order churn	
OPTIONS PRICING ENGINE - VANILLA, AMERICAN AND BARRIER	10/25
<i>C++20, CMake, CTest</i> github.com/tejaspandya9598/options-pricing-engine	
- Implemented three independent valuation methods behind one interface: closed-form Black-Scholes with analytic greeks, Monte Carlo with exact terminal sampling, and a Crank-Nicolson solver using projected SOR for American early exercise, with barriers as domain boundaries - Cross-validated Black-Scholes, Monte Carlo and Crank-Nicolson to within 0.6% on an at-the-money European call, pinning that agreement as regression tests at \$0.02 and \$0.06 tolerances, recovering a 0.51 early-exercise premium and documenting the discrete-monitoring gap on barriers	